

PARAMETER REVERSION

1. FROM **Q** PARAMETERS TO **P** MODELS

To the extent we trust our sense of what parameters in a model are reasonable, “stress” in model parameters correlates to profit opportunity. How do we decide when parameters are truly stressed?

2. PARAMETER CONES

Burghardt (1990) pointed out an elegant visual means to measure current parameter values of a parameter (in his case, volatility) against past quantiles.

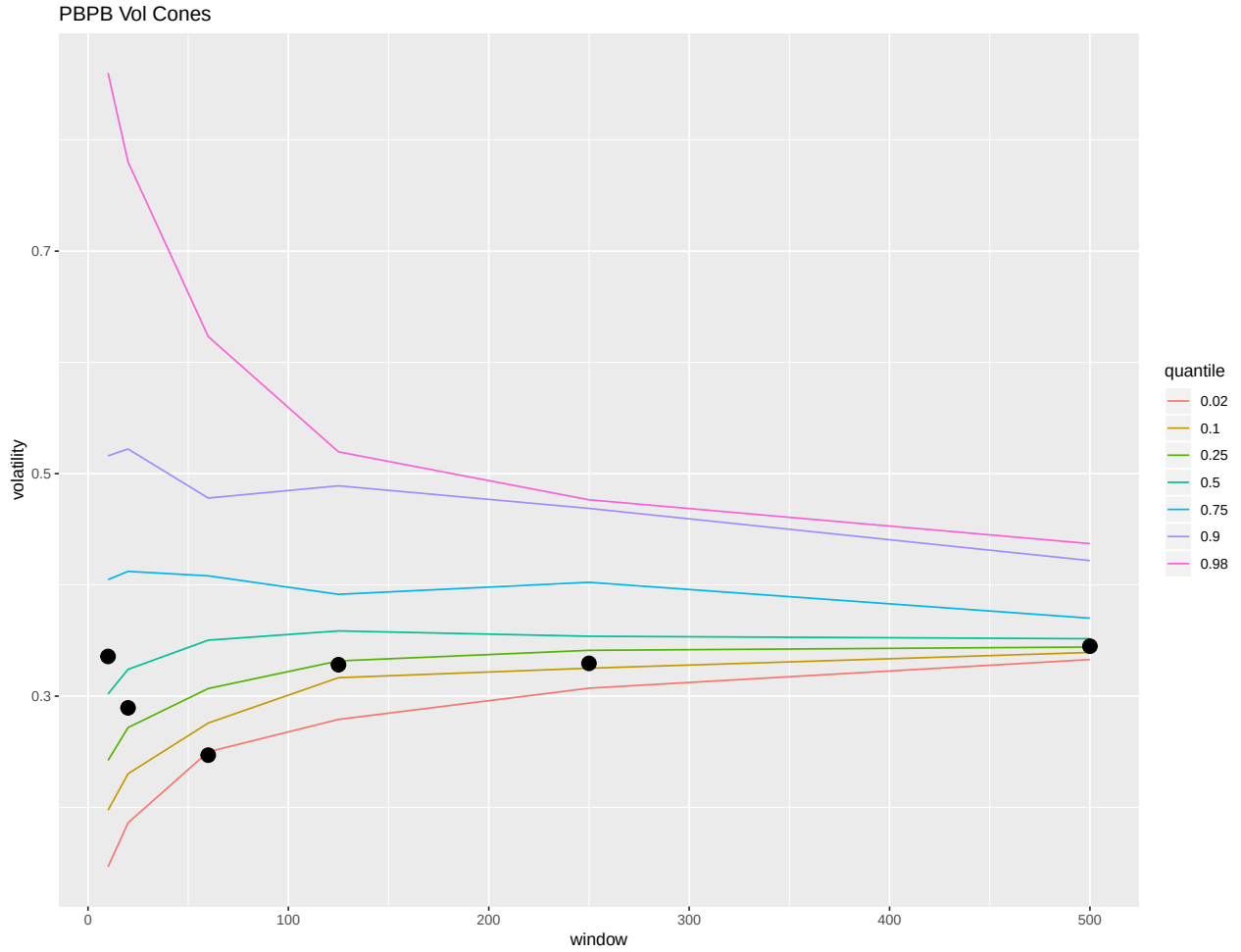


FIGURE 1. Volatility cones show recent vols are fairly normal, except perhaps at the 60 day scale.

3. TAXONOMY OF REVERSION

Parameters that tend to revert include (some) 10-Q financial accounting ratios, volatility surfaces and varvol¹ (Otherwise known as volatility of volatility.), yield curve shapes, default rates and risks, and general correlation.

Parameters that revert, but not reliably, include a few other financial accounting ratios, FX fundamentals, betas and pairwise correlations, and post-shock prices.

4. THE IMPLIED TRADING MODEL

The trading model we are inclined to choose based on parameter reversion is dependent, of course, on what assets have prices most sensitive to the parameter in question. It does not help to trade a lot of interest rate caplets if the curve parameter we are most sensitive to is yield curve tilt. For yield curve tilt would be better off with a spread option on different tenors. Similarly, if (as here) volatility is the parameter in question, we want our positions to express the expected change in volatility so they should have a high “vega” which in turn means they should be at-the-money or close to it.

The general principle is to find instruments with sufficient liquidity that are as sensitive as possible to changes in the parameter value. If we like, we can turn the parameter value stresses into predicted alphas, but this can interfere with performance if trading frequencies are low and/or costs are high.

5. BEWARE OF REGIME SHIFTS

The less well a model fits the market, and the more parameters it has, the likelier it is to fall victim to apparent changes in regime that have no economic basis. This can lead to an optimization algorithm finding jumps in model parameters that ultimately lead to noise trading or worse.

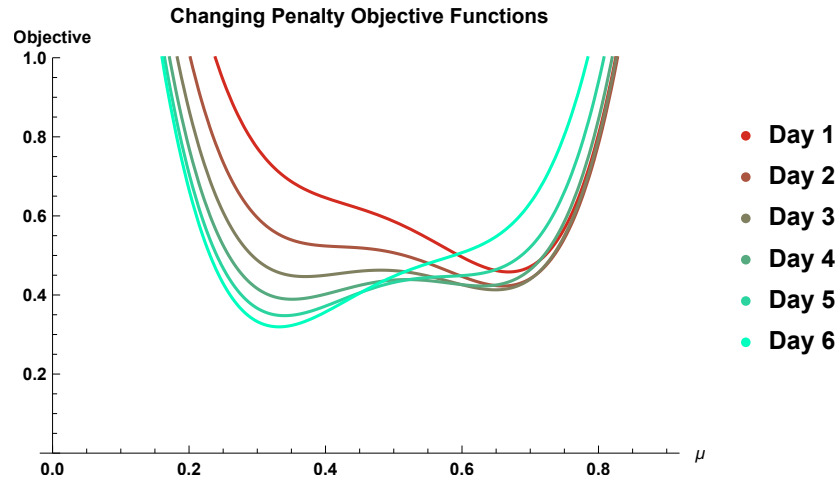


FIGURE 2. An objective function used for parameter choice may change as time passes and incoming information is updated.

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6. THE USES OF DECAYING WEIGHTS

Traditional statistics and parameter estimation from historical data have tended to use data windows (rolling ones if necessary). We see this in Burghardt's original definition of volatility cones.

In many cases, however, it is more reasonable to use some weighting scheme, where the weights decay toward zero as data items age. If we choose particular functional forms, particularly exponential weighting, computation costs can be greatly reduced as well.

The simplest example of this is the exponentially-weighted moving average where we take some $0 < \lambda < 1$ and compute the average

$$\bar{x} = \frac{\sum_{n=0}^{\infty} \lambda^n x_{-n}}{\sum_{n=0}^{\infty} \lambda^n}$$

This doesn't necessarily look cheap or simple, but it enjoys a 1-pass or *online algorithm* where we take (now for increasing n)

$$\bar{x}_0 = x_0$$

and

$$\bar{x}_{n+1} = \lambda \bar{x}_n + (1 - \lambda)x_{n+1}$$

Online algorithms, also known as *update algorithms* are algorithms in which a small amount of state information is stored, and each incoming datum allows us to derive the new desired output, and new algorithm state, in relatively few computations.

In the case of the EWMA, this state is simply the EWMA value $\bar{x}$ itself. For exponentially-weighted (or even windowed) estimates of variance, the state is actually two data elements corresponding roughly to the sum of squares and sum of square differences.

Many stochastic control perspectives and machine learning algorithms make heavy use of update algorithms.